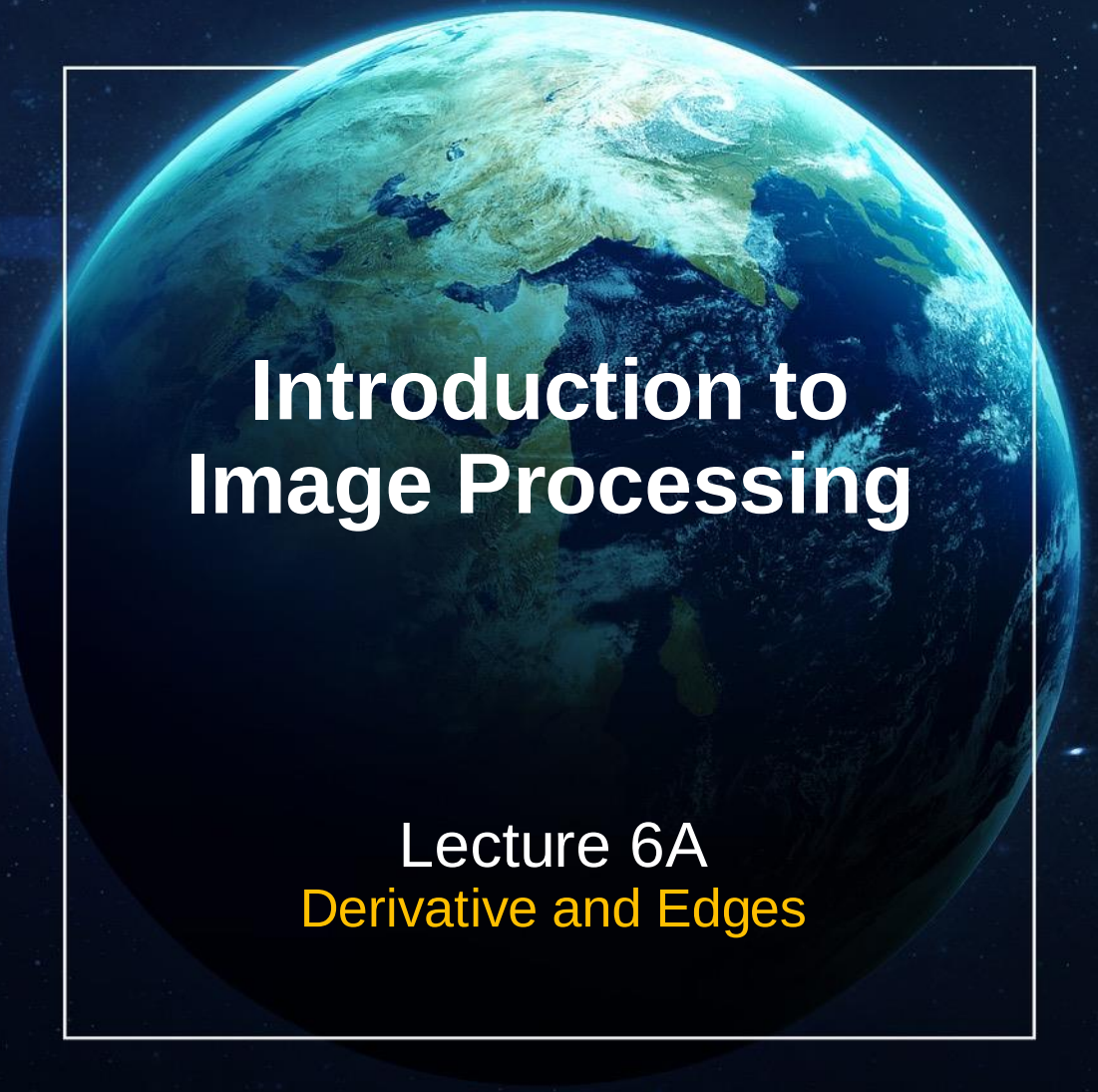




University of
Nottingham

UK | CHINA | MALAYSIA

A large, high-resolution image of the Earth as seen from space, showing the Western Hemisphere. The Earth is a vibrant blue and green sphere against a dark, star-filled background. A thin white rectangular border is superimposed over the center of the Earth.

Introduction to Image Processing

Lecture 6A
Derivative and Edges



Learning Outcomes

IDENTIFY

1. Derivative Filters
2. Sharpening
3. What is Edge Detection?
4. Edge Detection using 1st Derivatives
5. Edge Detection using 2nd Derivatives
6. The Canny Operator





University of
Nottingham

UK | CHINA | MALAYSIA

Derivative Filters



In 1 Dimension

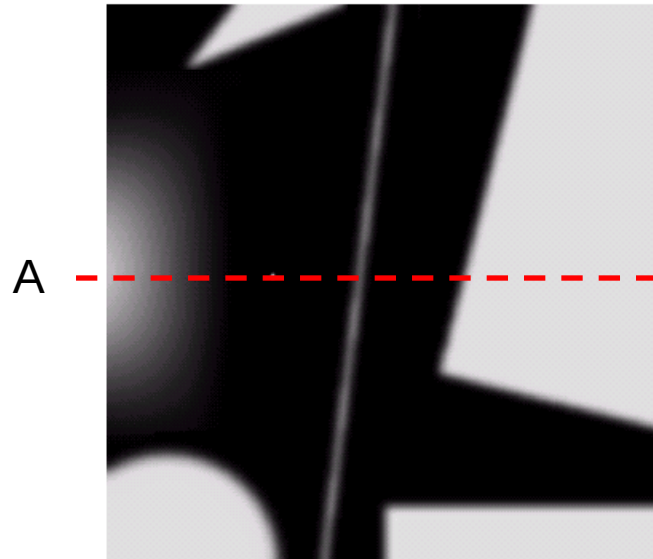
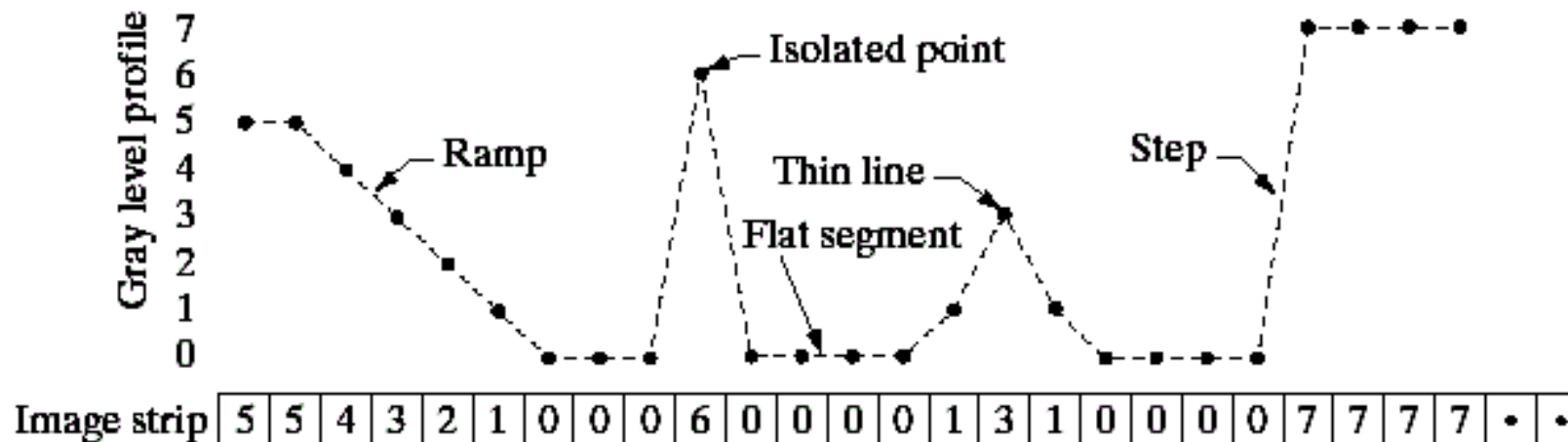


Image features are often characterised by changes in intensity





1st Derivative

The 1st derivative of a function can be approximated by:

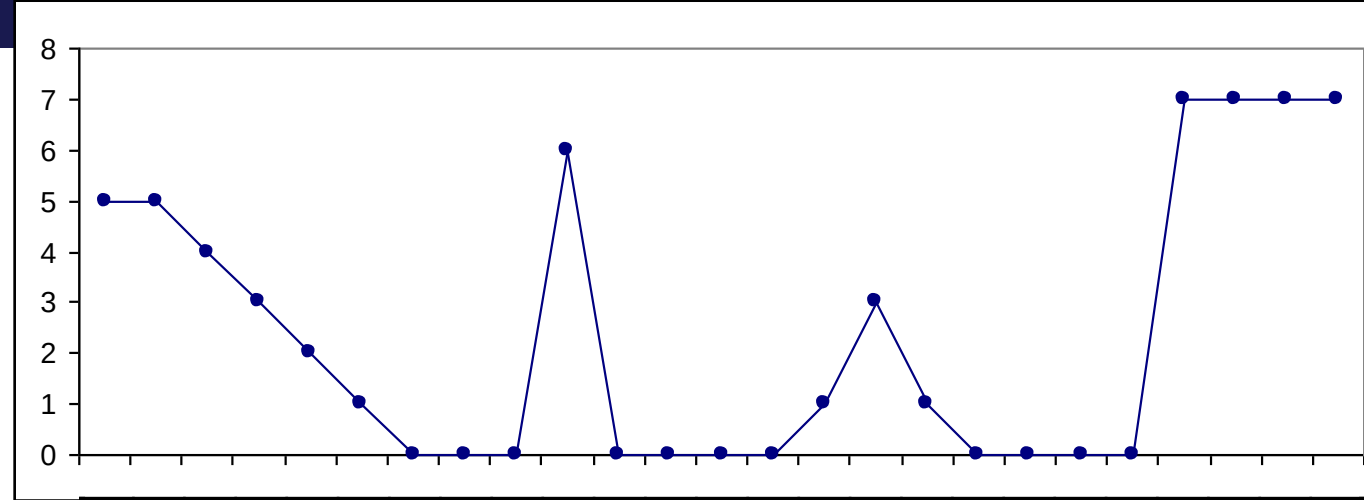
$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

The difference between neighbouring values and measures the rate of change of the function



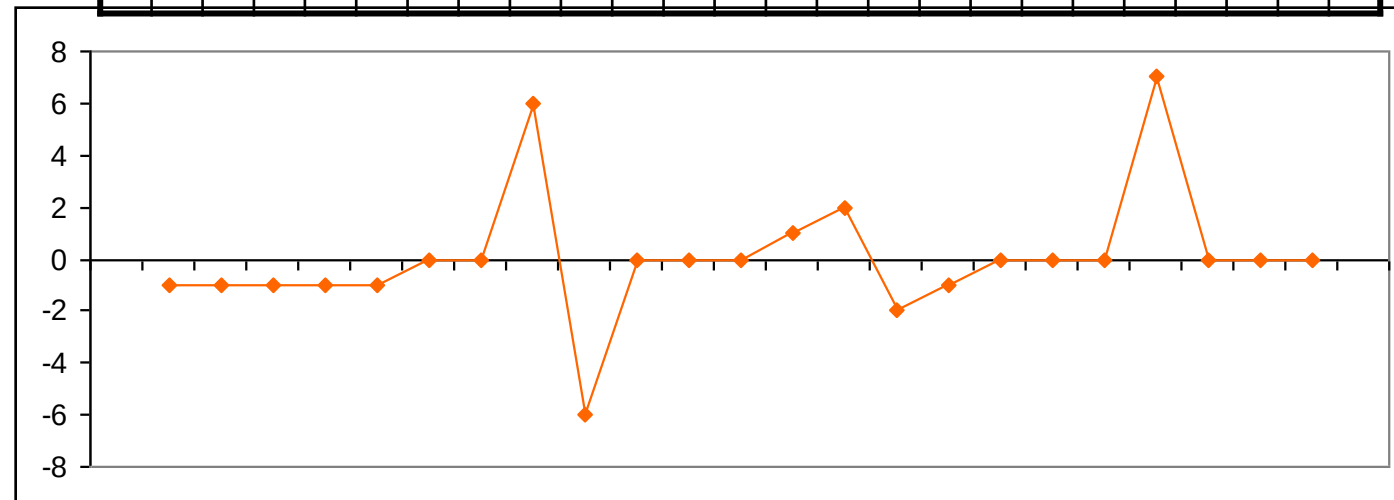
1st Derivative

Raw data



1st Derivative

	0	-1	-1	-1	-1	-1	0	0	6	-6	0	0	0	0	1	2	-2	-1	0	0	0	7	0	0	0
--	---	----	----	----	----	----	---	---	---	----	---	---	---	---	---	---	----	----	---	---	---	---	---	---	---





2nd Derivative

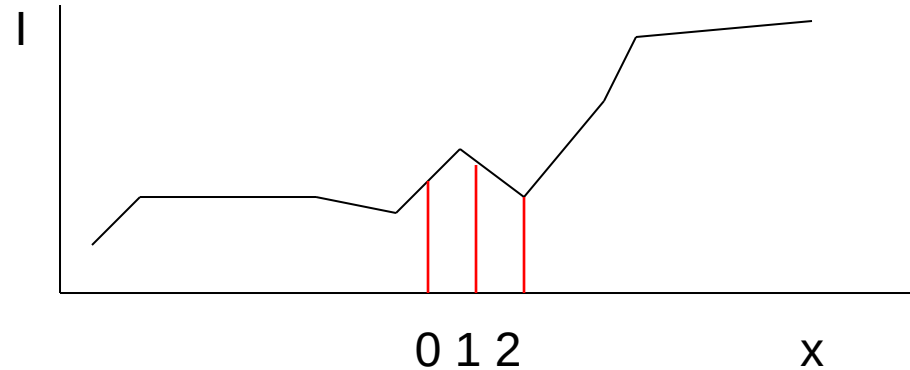
The formula for the 2nd derivative of a function is:

$$\frac{\partial^2 f}{\partial^2 x} = f(x+1) + f(x-1) - 2f(x)$$


- Simply takes into account the values of both – before and after the current value
- Derived by estimating the 1st derivative at $x + 0.5$ and $x - 0.5$ and computing the derivative of the resulting data



2nd Derivative



$$I''(1) = (I'(1.5) - I'(0.5))/1$$

$$I'(0.5) = (I(1) - I(0))/1 \quad \text{and} \quad I'(1.5) = (I(2) - I(1))/1$$

$$\therefore I''(1) = 1 \cdot I(0) - 2 \cdot I(1) + 1 \cdot I(2)$$

1

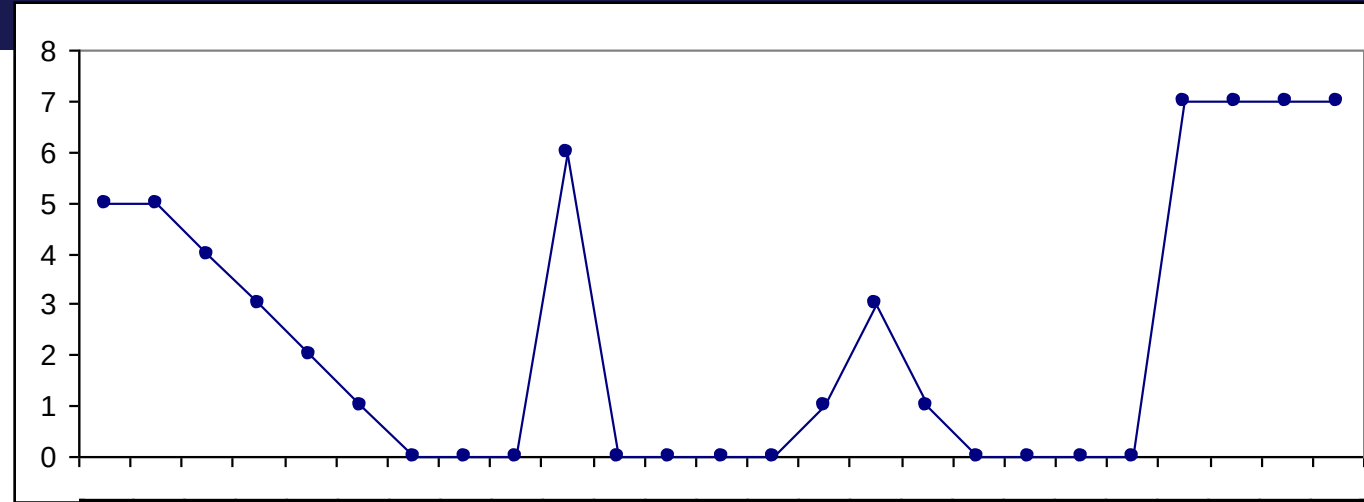
-2

1

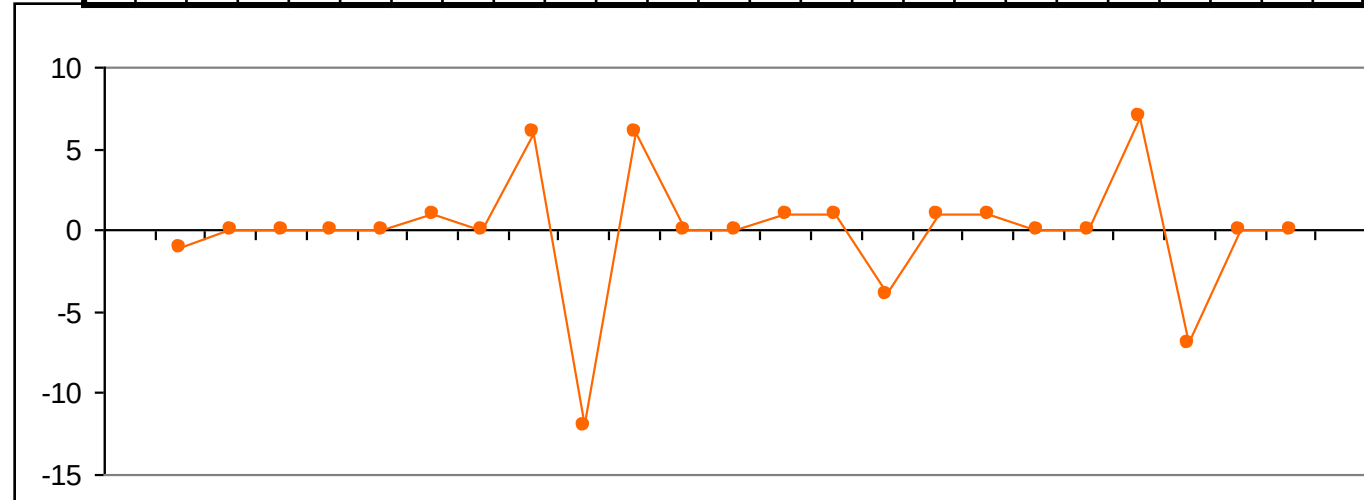
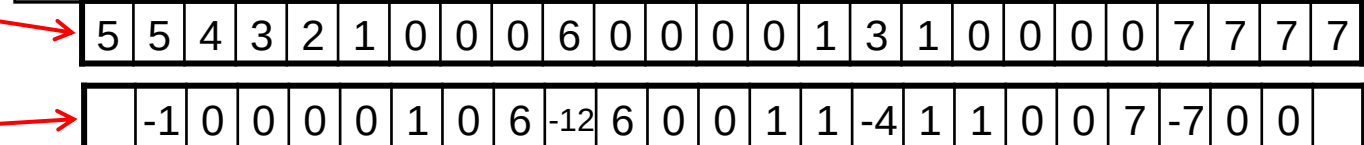


2nd Derivative

Raw data



2nd Derivative

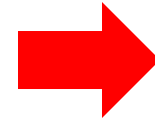




Derivatives in 2 Dimension

- 2nd derivatives generalise to 2D quite easily, implementing a 1st derivative in 2D is a little more complex

For a function $f(x,y)$ the gradient of f at coordinates (x,y) is given as a column vector:



$$\nabla f = \begin{bmatrix} G_x \\ G_y \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

REMEMBER

- Computation of the 1st derivative can't be done by convolution alone



1st Derivative Filtering

The magnitude of the 1st derivative vector is

$$\nabla f = \text{mag}(\nabla f)$$

$$= [G_x^2 + G_y^2]^{1/2}$$

$$= \left[\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right]^{1/2}$$

which can be simplified to

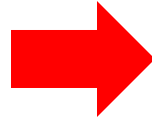
$$\nabla f \approx |G_x| + |G_y|$$



1st Derivative Filters

Many 1st derivatives filters have been proposed

Roberts' Cross Operators



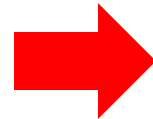
1	0
0	-1

G_x

0	1
-1	0

G_y

Sobel Operators



-1	0	1
-2	0	2
-1	0	1

G_x

-1	-2	-1
0	0	0
1	2	1

G_y

These operators are most commonly associated with edge detection



University of
Nottingham

UK | CHINA | MALAYSIA

Image Sharpening



Edge Enhancement: Unsharp Masking

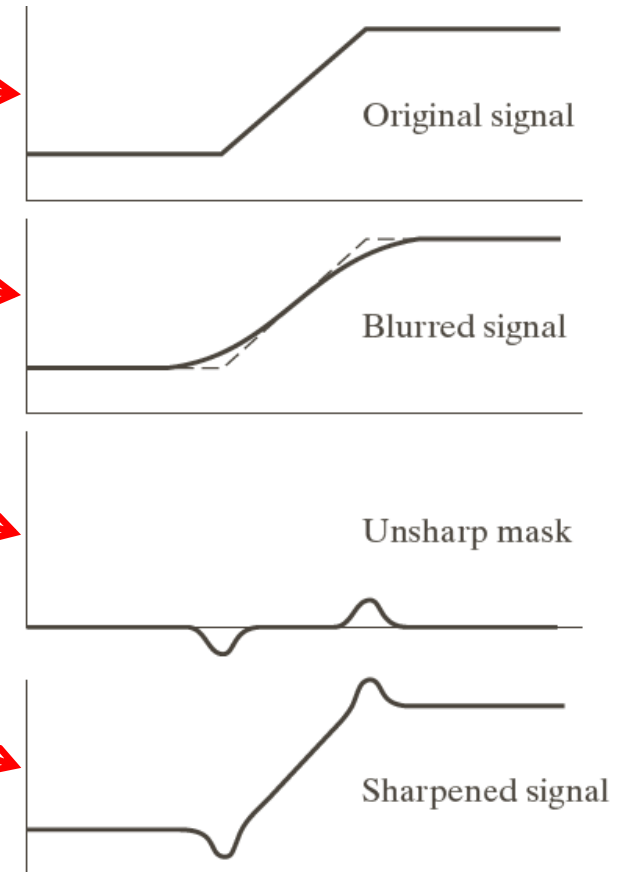
- Edges are important
- Sometimes we want to enhance them without (much) affecting the rest of the image

1. Take the original image

2. Gaussian smooth it

3. Subtract the smoothed version from the original to make an *unsharp mask*

4. Add the mask to the original to make the edge appear more obvious

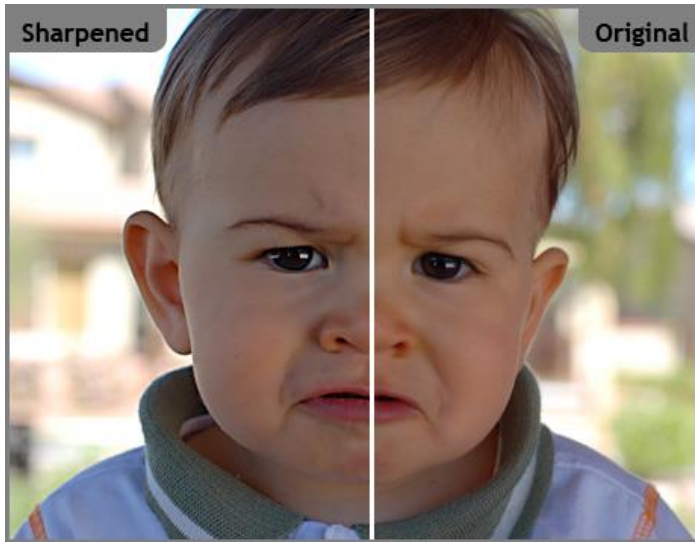




Unsharp Masking

Makes edges noticeable sharper

- Even if they are noise
- Sometimes too much





Derivative Filters

Unsharp filtering enhances edges by comparing the original with a smoothed image

- Relies on the smoothing effect of a Gaussian function introducing a difference between original and processed images
- Parameterised by σ
- Simple, but effect is hard to predict, so hard to parameterise

A more direct way to highlight edges and other features associated with high image gradients is to estimate derivatives...



Image Sharpening with Derivatives

The 2nd derivatives is more useful for image enhancement than the 1st derivative

- Stronger response to fine detail
- Simpler implementation

The most common sharpening filter is the *Laplacian*

- Isotropic
- One of the simplest sharpening filters
- Straightforward digital implementation via convolution



The Laplacian

$$\nabla^2 f = \frac{\partial^2 f}{\partial^2 x} + \frac{\partial^2 f}{\partial^2 y}$$

$$\frac{\partial^2 f}{\partial^2 x} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial^2 y} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$



The Laplacian

$$\nabla^2 f = [f(x+1, y) + f(x-1, y) \\ + f(x, y+1) + f(x, y-1)] \\ - 4f(x, y)$$

0	1	0
1	-4	1
0	1	0

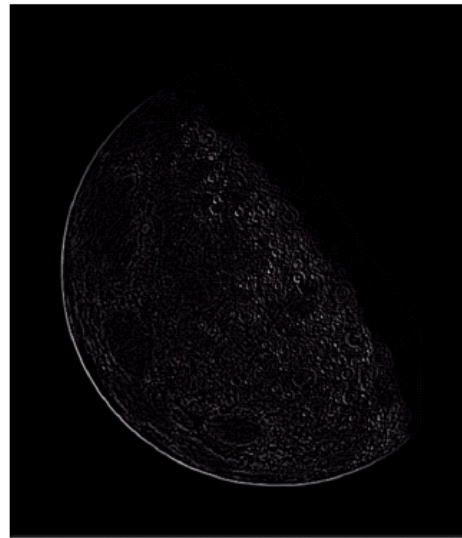


The Laplacian

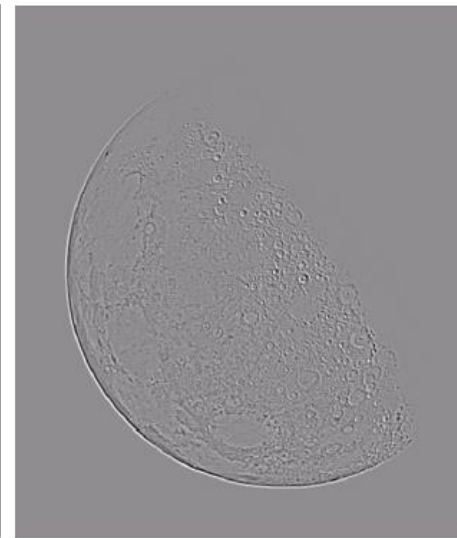
Highlights edges and other discontinuities



Original
Image



Laplacian
Filtered Image



Laplacian
Filtered Image
Scaled for Display



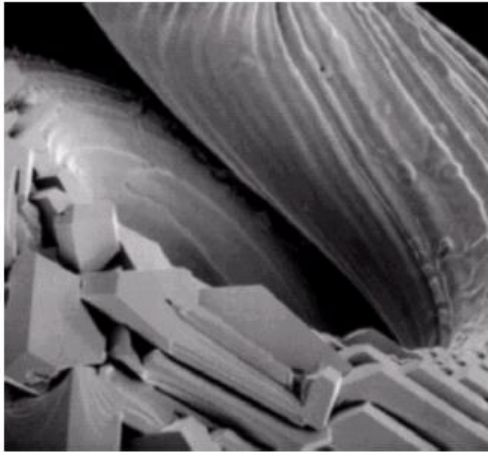
A Single Enhancement Operator

$$\begin{aligned} g(x, y) &= f(x, y) - \nabla^2 f \\ &= f(x, y) - [f(x+1, y) + f(x-1, y) \\ &\quad + f(x, y+1) + f(x, y-1) \\ &\quad - 4f(x, y)] \\ &= 5f(x, y) - f(x+1, y) - f(x-1, y) \\ &\quad - f(x, y+1) - f(x, y-1) \end{aligned}$$



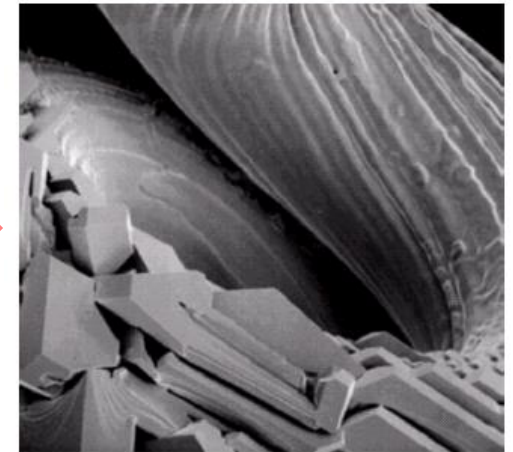
A Single Operator

Convolution with this operator performs image sharpening in a single step



Input

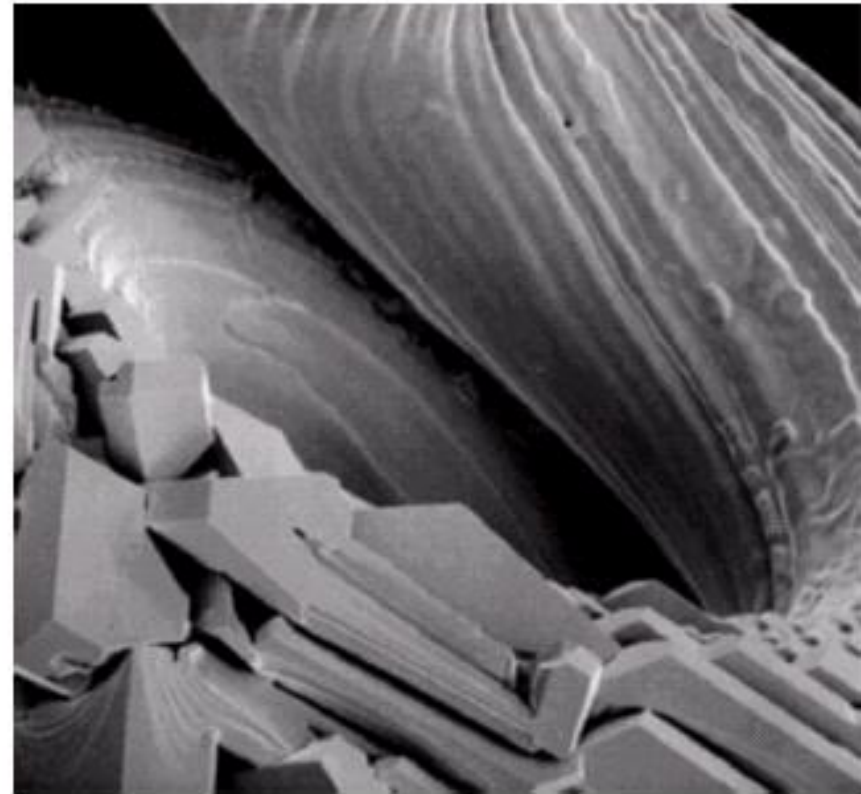
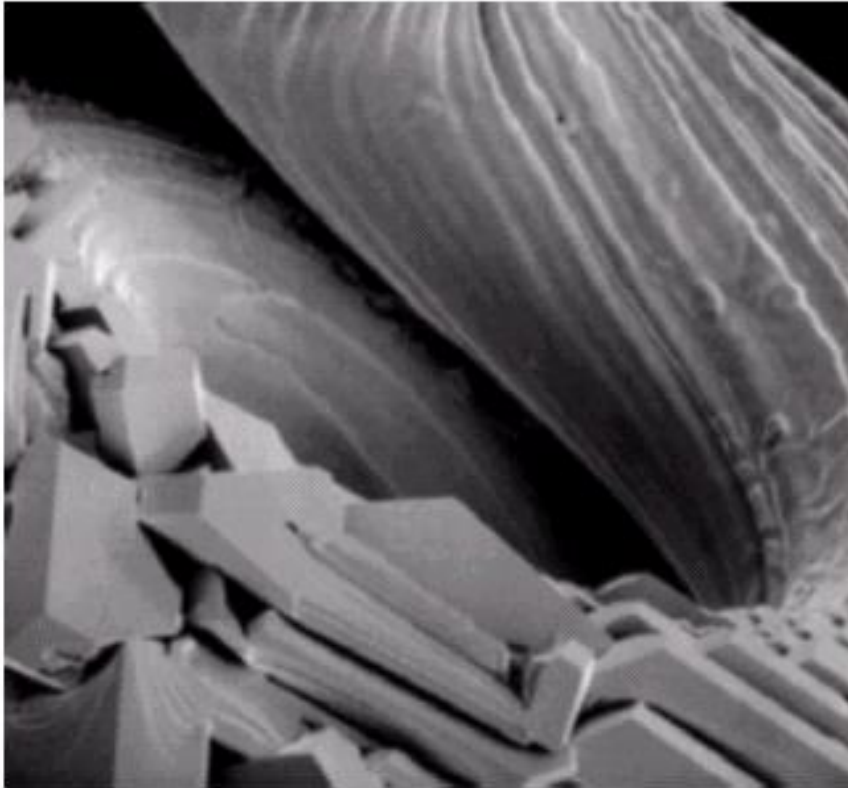
0	-1	0
-1	5	-1
0	-1	0



Sharpen



A Single Operator



Let's take a
CLOSER look



Variations on the Theme

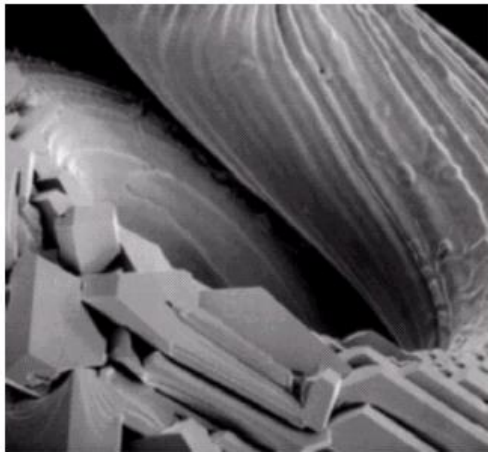
0	1	0
1	-4	1
0	1	0

Basic

Some formulations take 2nd derivatives measured across the diagonals into account

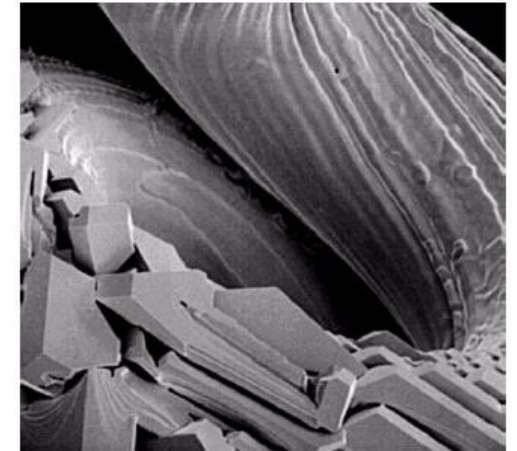
1	1	1
1	-8	1
1	1	1

Extended



Input

-1	-1	-1
-1	9	-1
-1	-1	-1



Sharpen



University of
Nottingham

UK | CHINA | MALAYSIA

What is **Edge Detection?**

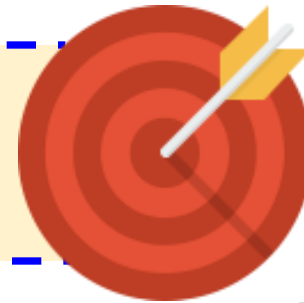


Edge Detection

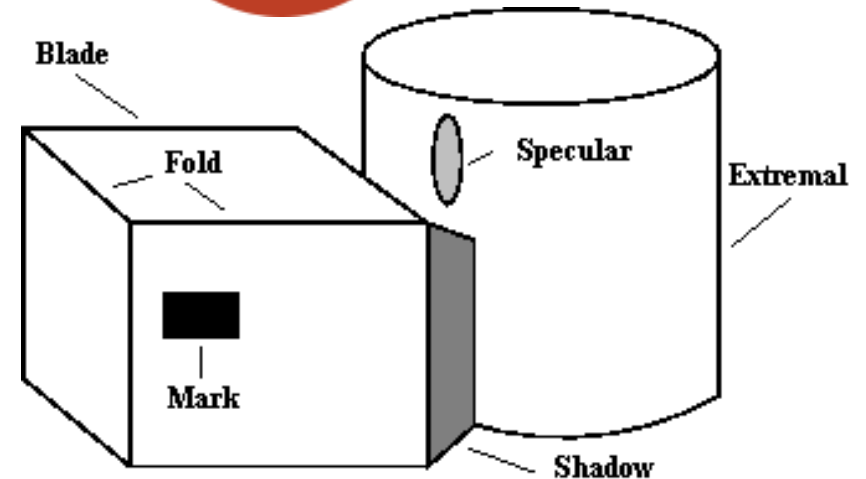
First Step

In many image analysis and computer vision processes and applications

To mark points at which image intensity changes sharply - **edges**

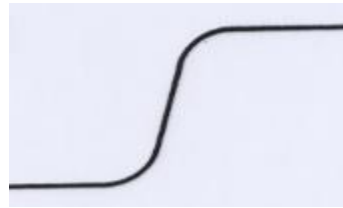
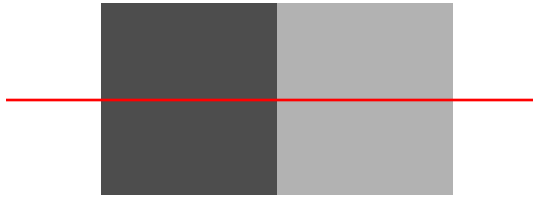


- Sharp changes in **image** properties reflect events/changes in the **world**
- This is only an assumption, but it is usually true

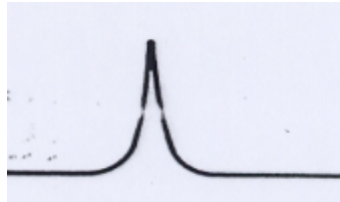




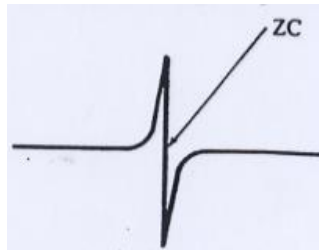
The Theory



An idealised
image of an edge



1st derivative: **a peak**



2nd derivative: **a zero-crossing**

To detect edges find peaks in the 1st derivative or intensity or zero-crossings in the 2nd derivative



The Result

```
>> im = imread('cameraman.tif');  
>> edges = edge(im, 'Canny');  
>> imshowpair(im, edges, 'montage');
```





University of
Nottingham

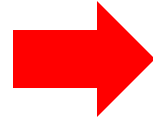
UK | CHINA | MALAYSIA

Edge Detection using **1st** **Derivative Filters**



1st Derivative Filters

Roberts' Cross Operators



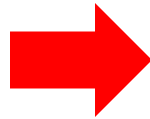
1	0
0	-1

G_x

0	1
-1	0

G_y

Sobel Operators



-1	0	1
-2	0	2
-1	0	1

G_x

-1	-2	-1
0	0	0
1	2	1

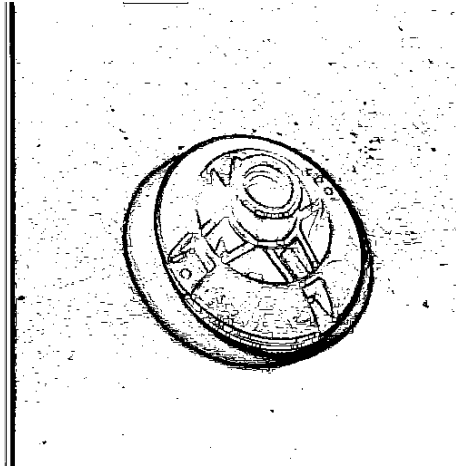
G_y

Applied separately and results combined to estimate magnitude

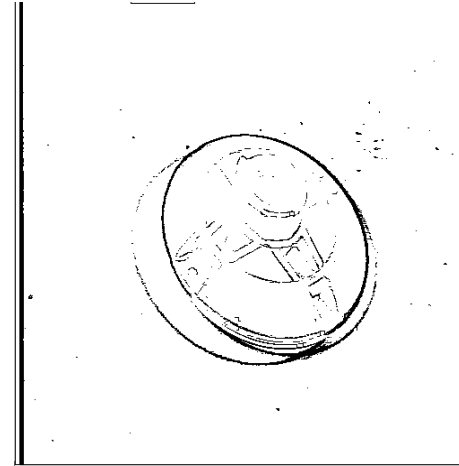


Detection & Thresholding

- Significant peaks in magnitude of 1st derivative are high
- Apply a threshold, all peaks higher than the threshold value are significant, all other are ignored



Too low



Too high



Edge Magnitude & Direction

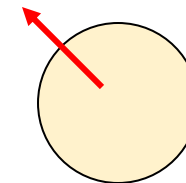
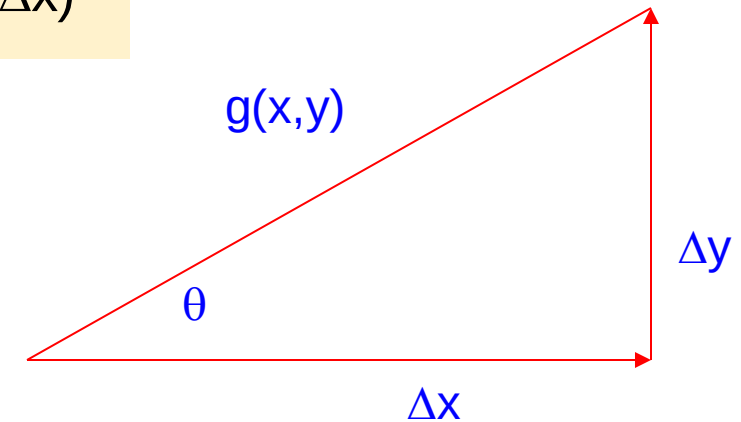
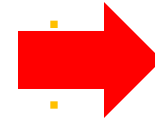
For an image function, $I(x,y)$, the gradient magnitude, $g(x,y)$ is given by

$$g(x,y) \cong (\Delta x^2 + \Delta y^2)^{1/2}$$

The gradient direction, $\theta(x,y)$, gives the direction of steepest image gradient

$$\theta(x,y) \cong \text{atan}(\Delta y / \Delta x)$$

This gives the direction of a line perpendicular to the edge



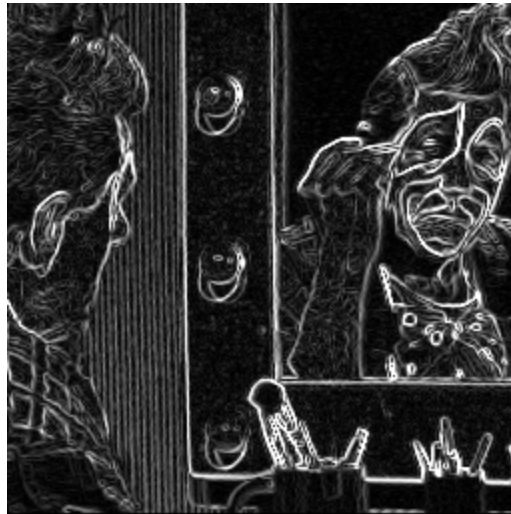


Roberts' Cross Operator

Very quick to compute – 4 pixels, only subtractions and additions, **but is very sensitive to noise** and only gives a strong response to very sharp edges



Original



Cross Operator



Thresholded

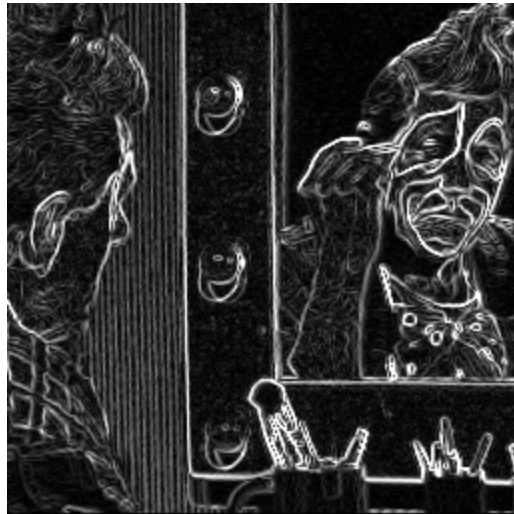


Sobel **vs** Roberts

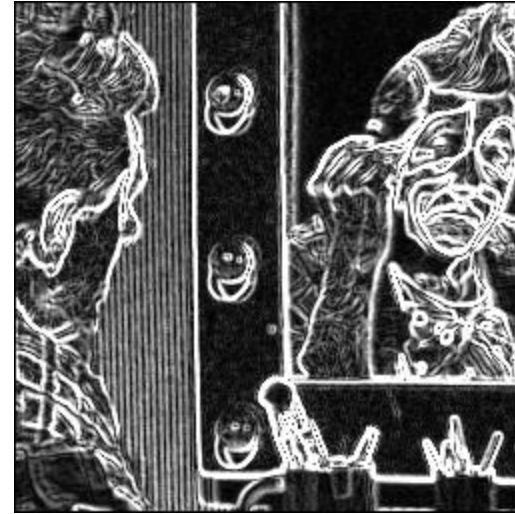
- Both use a super-supplied threshold. Sobel is still in use. Roberts is less common, nowadays.
- Larger Sobel operators are more stable in noise



Original



Roberts



Sobel



University of
Nottingham

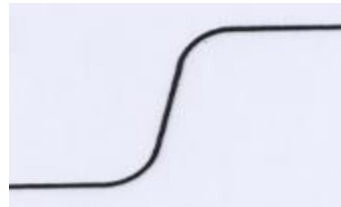
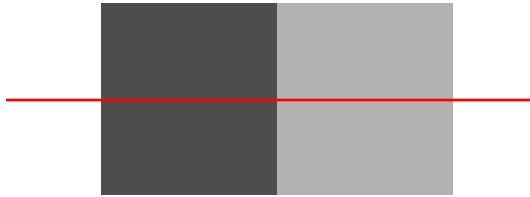
UK | CHINA | MALAYSIA

Edge Detection using **2nd** **Derivative Filters**

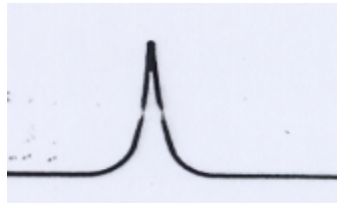


The Theory

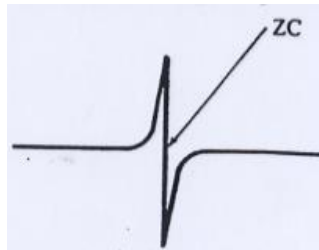
QUICK RECAP!



An idealised
image of an edge



1st derivative: **a peak**



2nd derivative: **a zero-crossing**

To detect edges find peaks in the 1st derivative or intensity or zero-crossings in the 2nd derivative



2nd Derivatives: Marr-Hildreth

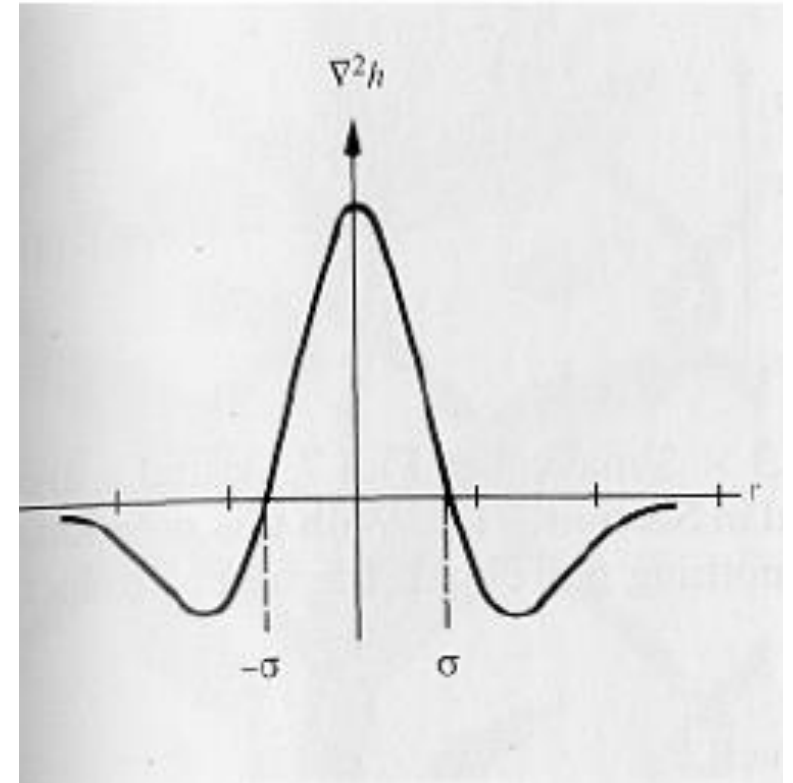
Biologically inspired

Gaussian smooth, compute Laplacian

OR

Convolve with the **Laplacian of Gaussian**

$$\nabla^2[f(x, y) * G(x, y)] = \nabla^2 G(x, y) * f(x, y)$$





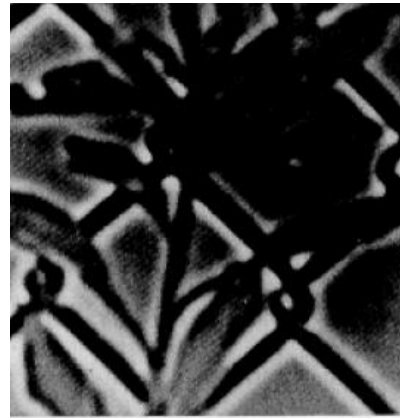
Laplacian of Gaussian (LoG)

5 × 5 Laplacian of Gaussian mask

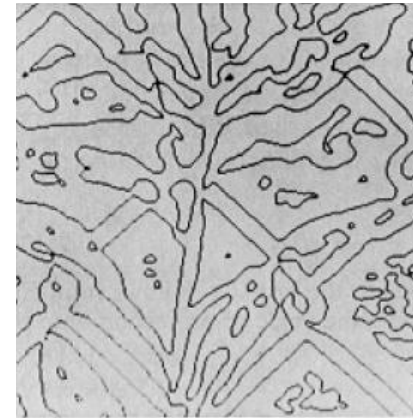
0	0	-1	0	0
0	-1	-2	-1	0
-1	-2	16	-2	-1
0	-1	-2	-1	0
0	0	-1	0	0

17 × 17 Laplacian of Gaussian mask

0	0	0	0	0	0	-1	-1	-1	-1	-1	0	0	0	0	0	0
0	0	0	0	-1	-1	-1	-1	-1	-1	-1	-1	-1	0	0	0	0
0	0	-1	-1	-1	-2	-3	-3	-3	-3	-3	-2	-1	-1	-1	0	0
0	0	-1	-1	-2	-3	-3	-3	-3	-3	-3	-3	-2	-1	-1	0	0
0	-1	-1	-2	-3	-3	-3	-2	-3	-2	-3	-3	-3	-2	-1	-1	0
0	-1	-2	-3	-3	-3	0	2	4	2	0	-3	-3	-3	-2	-1	0
-1	-1	-3	-3	-3	0	4	10	12	10	4	0	-3	-3	-3	-1	-1
-1	-1	-3	-3	-2	2	10	18	21	18	10	2	-2	-3	-3	-1	-1
-1	-1	-3	-3	-3	4	12	21	24	21	12	4	-3	-3	-3	-1	-1
-1	-1	-3	-3	-2	2	10	18	21	18	10	2	-2	-3	-3	-1	-1
-1	-1	-3	-3	-3	0	4	10	12	10	4	0	-3	-3	-3	-1	-1
0	-1	-2	-3	-3	-3	0	2	4	2	0	-3	-3	-3	-2	-1	0
0	-1	-1	-2	-3	-3	-3	-2	-3	-2	-3	-3	-3	-2	-1	-1	0
0	0	-1	-1	-2	-3	-3	-3	-3	-3	-3	-3	-2	-1	-1	0	0
0	0	-1	-1	-1	-2	-3	-3	-3	-3	-3	-2	-1	-1	-1	0	0
0	0	0	0	-1	-1	-1	-1	-1	-1	-1	-1	-1	0	0	0	0
0	0	0	0	0	0	-1	-1	-1	-1	-1	0	0	0	0	0	0



Filtering



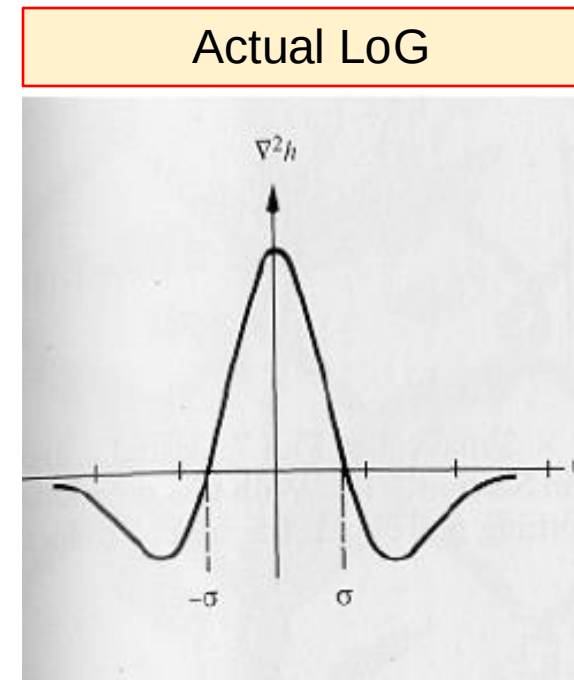
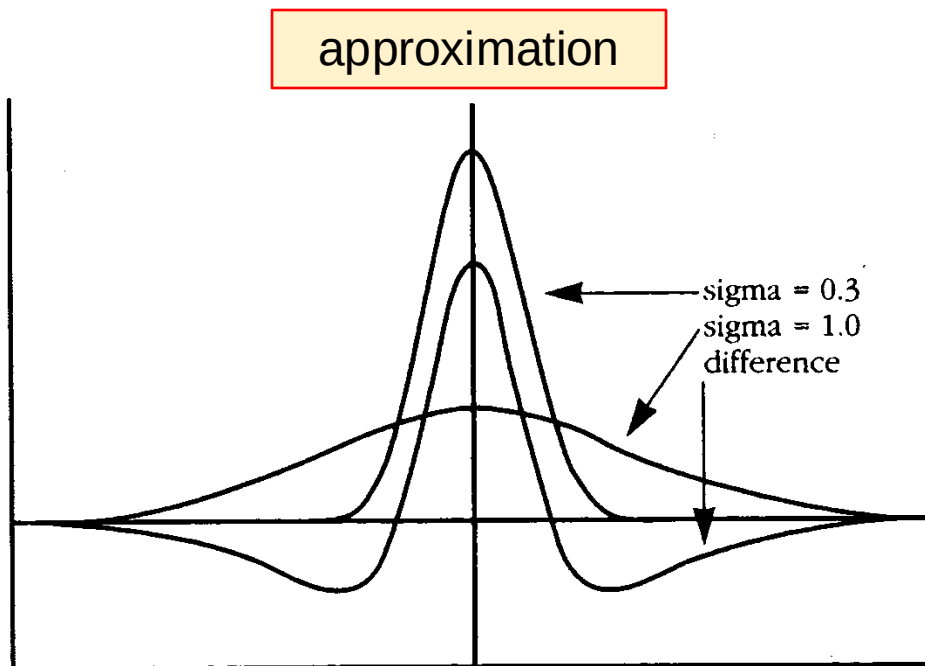
Zero-Crossings



Difference of Gaussians

The **Laplacian of Gaussian** can be approximated by the difference between two Gaussian functions:

$$\nabla^2 G \approx G(x, y; \sigma_1) - G(x, y; \sigma_2)$$





DoG Filtering

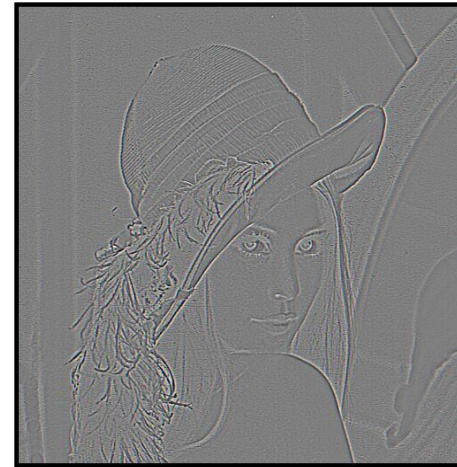
$$\nabla^2 G \approx G(x, y; \sigma_1) - G(x, y; \sigma_2)$$



(a) $\sigma = 1$



(b) $\sigma = 2$



(b) - (a)

Ratio (σ_1/σ_2) for best approximation is about 1.6.
(Some people like $\sqrt{2}$.)

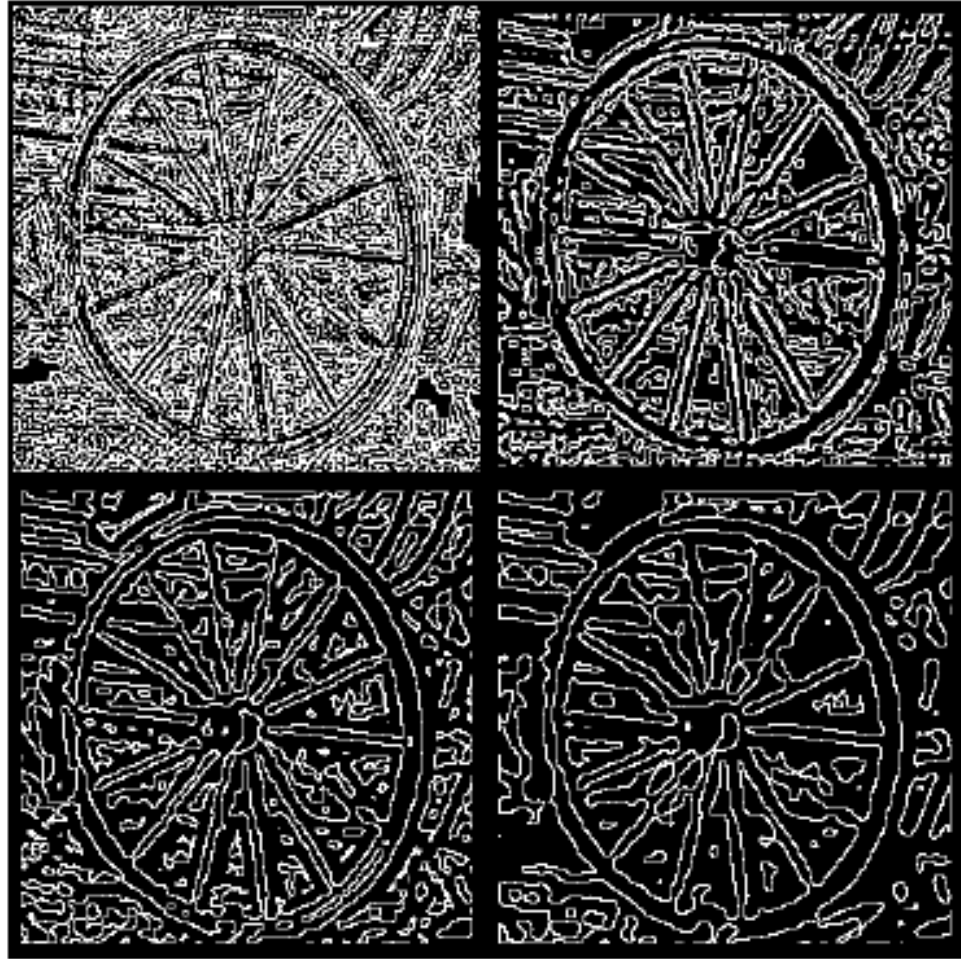
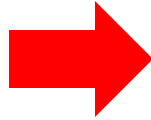


Marr-Hildreth

Choice of σ gives flexibility



Input Image



$\sigma = 1$

$\sigma = 2$

$\sigma = 3$

$\sigma = 4$



1st vs 2nd Derivative Methods

Peaks in 1st Derivative

- Strong response at edges, but also respond to noise
- Peak detection and threshold selection need care

VS

Zero crossings in 2nd derivative

- Well-defined, easy to detect
- Must form smooth, connected contours
- Tend to round off corners

1st derivative methods are much more common in practical applications,

In part because of John Canny



University of
Nottingham

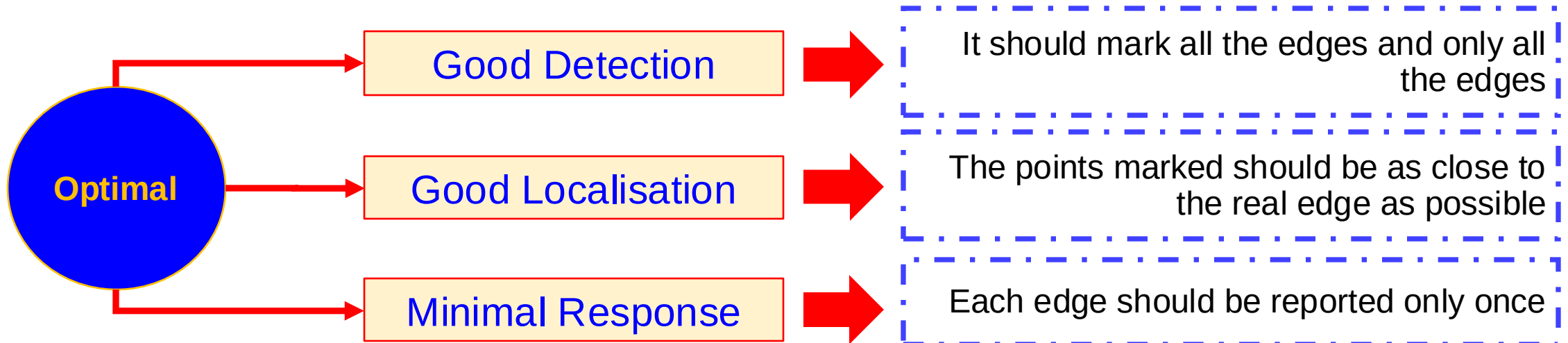
UK | CHINA | MALAYSIA

The Canny Operator



What Canny Did

John Canny tried to find the optimal edge detector, assuming a perfect step edge in Gaussian noise



Canny used the *Calculus of Variations*: finds the function which best satisfies some functional

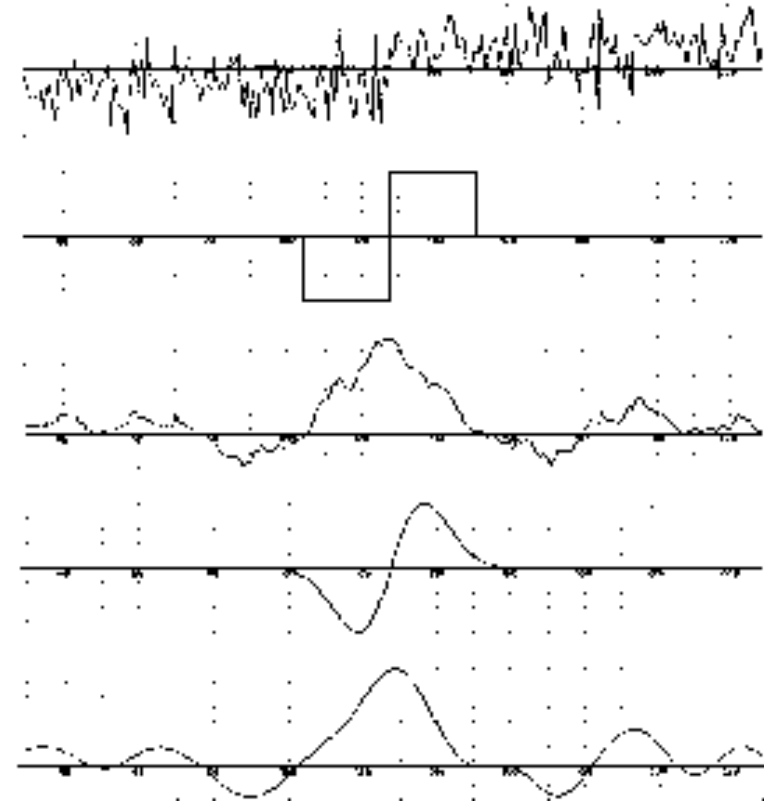


The Canny Operator

- The optimal detector was a sum of 4 exponential terms, but is very closely approximated by the 1st derivative of a Gaussian

i.e., 1st derivative of a Gaussian smoothed image

- Gives a cleaner response to a noisy edge than square operators
- Most implementations are 2D Gaussian smoothing + Roberts style derivative



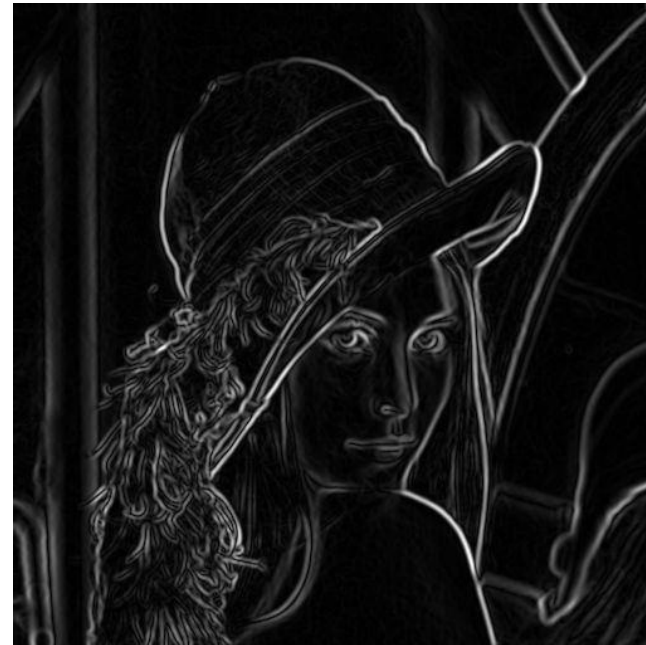
Roberts

Canny



Non-Maximal Suppression

The Canny operator's response is cleaner than Sobel or Roberts, but it needs an explicit step to enforce Minimal Response



Canny operator



Non-Maximal Suppression

Thresholding raw operator response would leave thick lines

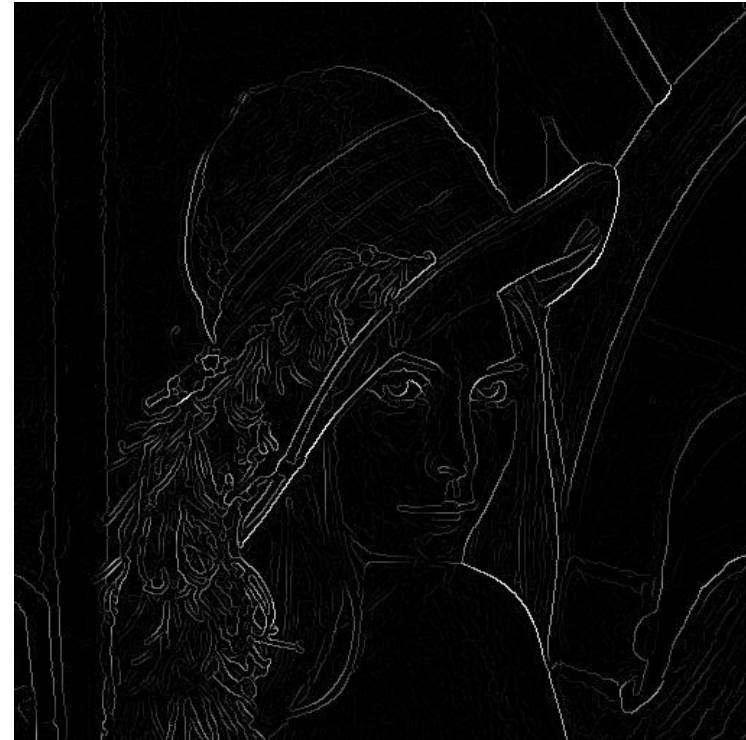
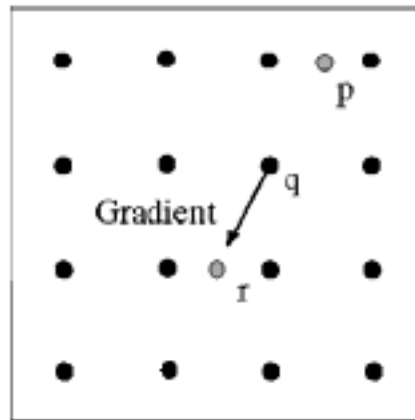
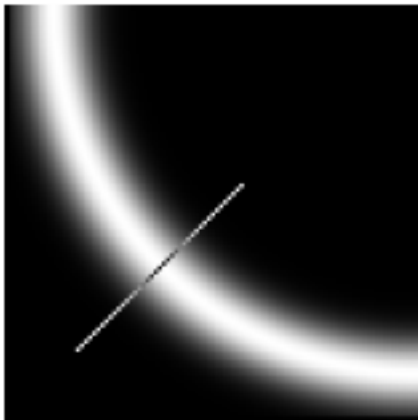


How to turn these thick regions of the gradient into curves?



Non-Maximal Suppression

1. Check if pixel is a local maximum along the gradient direction
2. Select a single maximum across the width of the edge

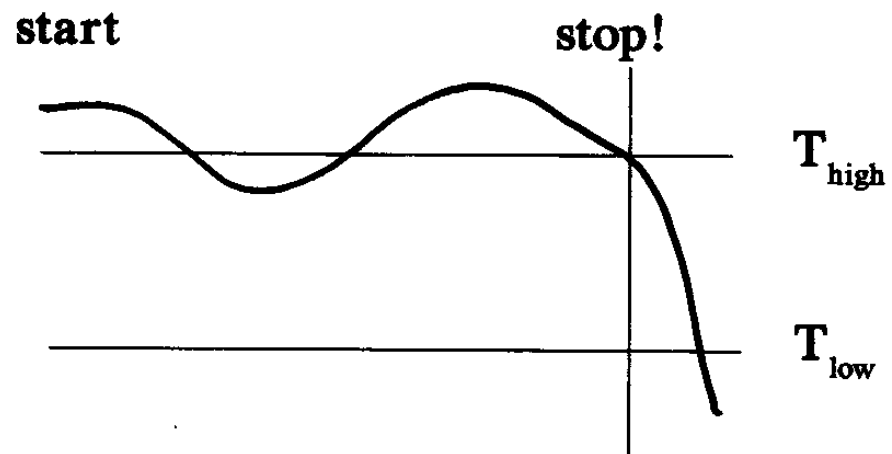




Thresholding with Hysteresis

- Simple thresholding tests each pixel independently:
edges aren't really independent, they make up lines
- The industry standard edge thresholding method

- Allows a band of variation, but assumes continuous edges
- User still selects parameters, but its easier, less precise

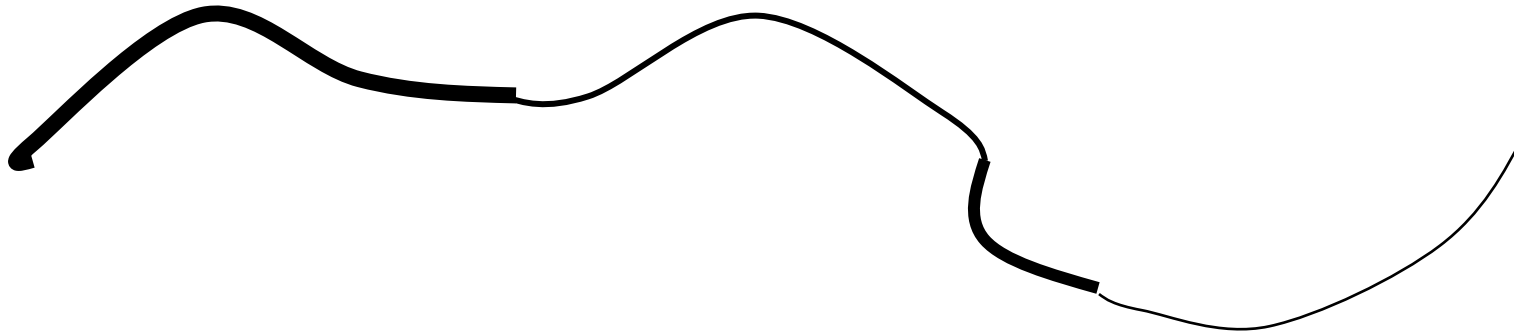




Thresholding with Hysteresis

The effect is to keep weak edges if they connect strong edges, as long as

- The strong edges are really strong
- The weak edges aren't really weak





Thresholding with Hysteresis

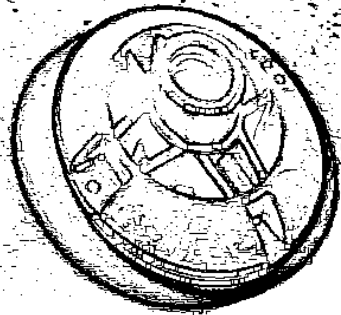
Hysteresis fills in most of the gaps



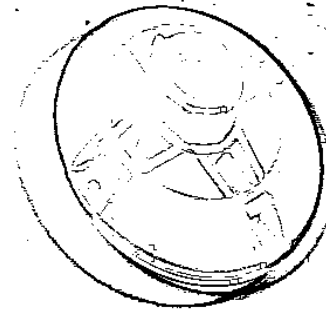
Problem:
pixels along
this edge
didn't
survive the
thresholding



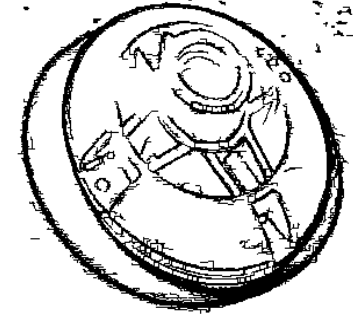
Thresholding with Hysteresis



Low threshold



High threshold



Hysteresis



What Canny Did

Showed that 1st derivative of a Gaussian smoothed image is the optimal way to detect step edges in noise

Explained why 1st derivatives are a good idea

Designed the industry standard thresholding method

- Non-maximal suppression
- Thresholding with hysteresis

Effectively **solved** the edge detection problem

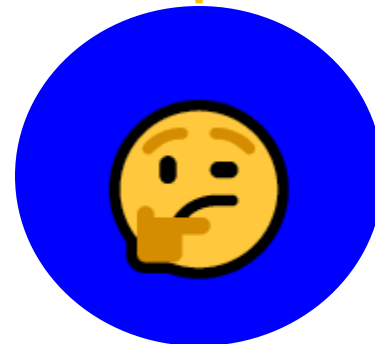




Summary



1. Derivative Filters
2. Sharpening
3. What is Edge Detection?
4. Edge Detection using 1st Derivatives
5. Edge Detection using 2nd Derivatives
6. The Canny Operator





University of
Nottingham

UK | CHINA | MALAYSIA

Questions



University of
Nottingham

UK | CHINA | MALAYSIA

NEXT:

Hough Transform